SafeRoute Gujarat: formula glossary

Every term in every formula on the presentation slides, as computed by the model. Slide 8: safety formulas · Slide 9: traffic-flow and scoring formulas · Slide 11: route cost.

Slide 8: Safety formulas

8.1

Severity (Nilsson power model)

How much harm a crash does at speed v, compared with a crash at 40 km/h.

$$S(v) = (1-\varphi)\left(\frac{v}{40}\right)^2 + \varphi\left(\frac{v}{40}\right)^4$$

Term	Meaning
$S(v)$	Severity multiplier. It equals 1 at 40 km/h and grows rapidly above that.
v	Operating speed in km/h: the road's speed limit (or a default for its road class), reduced for congestion at that hour. At junctions it is the fastest approaching road's speed, capped at 60 km/h.
40	Reference speed in km/h, a typical urban speed limit.
$\frac{v}{40}$	How many times faster than 40 km/h the vehicle is moving.
$\left(\frac{v}{40}\right)^2$	The injury part: harm in injury crashes grows with the square of speed.
$\left(\frac{v}{40}\right)^4$	The fatal part: the chance of death grows with the fourth power of speed.
φ (phi)	Share of crashes that are fatal, set to 0.33. MoRTH 2022 deaths per crash: Ahmedabad 0.29, Vadodara 0.38, Surat 0.43.
$1-\varphi$	Share of crashes that are non-fatal (0.67), which get the injury weighting.

Worked example (80 km/h): $0.67 \times 2^2 + 0.33 \times 2^4 = 2.68 + 5.28 \approx 8$, so about 8 times the harm at 40 km/h.

8.2 Lighting factor (Highway Safety Manual CMF)

How much an unlit road at night raises crash risk.

$$\mathrm{CMF}_{\mathrm{light}} = 1 + 0.39\ (1 - p_{\mathrm{lit}})\, d(h)$$

Term	Meaning
$\mathrm{CMF}_{\mathrm{light}}$	Crash Modification Factor for lighting. It multiplies a road's crash risk; 1 means no change.
1	Baseline: a fully lit road, or any road in daylight, gets no penalty.
0.39	From the Highway Safety Manual: lighting reduces night injury crashes by about 28%, so unlit ÷ lit = 1 ÷ 0.72 = 1.39, and the extra is 0.39.
p_{lit}	Probability that the road is lit (0 to 1): 75% satellite night-light brightness plus 25% a default for the road class. Overridden by OpenStreetMap: tagged lit gives 0.97, tagged unlit 0.03, and a mapped street lamp at least every 100 m gives at least 0.9.
$1 - p_{\mathrm{lit}}$	Probability that the road is dark.
$d(h)$	Darkness at departure hour h: 0 from 7 am to 6 pm, rising to 1 between 6 and 7:30 pm, 1 through the night, falling back to 0 between 6 and 7 am.
h	Hour of departure (0–24).

Example: a completely dark road ($p_{\mathrm{lit}} = 0$) at midnight ($d = 1$) gives CMF = 1.39. At noon $d = 0$, so the CMF is 1 for every road.

8.3 Turns at junctions

Each manoeuvre through a junction gets its own risk.

$$R_{\text{turn}} = k \cdot R_{\text{junction}}$$

Term	Meaning
R_{turn}	Risk of making one particular manoeuvre (left, straight, right or U-turn) at a junction.
R_{junction}	The junction's base risk: a calibration constant × an arms factor (3-way 1.0, 4-way 1.5, 5+ 1.8) × a control factor (signal 0.70, roundabout 0.60, stop/give-way 0.90, none 1.0) × (minor-road traffic ÷ 1000)$^{0.35}$ × the severity, time-of-day and lighting factors, × the blackspot multiplier if one is nearby.
k	Turn factor, chosen by the change in heading: straight (under 35°) 1.00; left turn 0.75; right turn 1.60, because it crosses oncoming traffic in left-hand driving; U-turn (over 155°) 2.20.

The idea that right turns are the riskiest comes from Indo-HCM conflict studies. The four k values are our modelling assumptions.

8.4 Empirical Bayes at blackspots

Blends a blackspot's crash record with the model's prediction.

$$w = \frac{1}{1 + 0.6\, N_{\text{pred}}\, Y}, \qquad N_{\text{EB}} = w\, N_{\text{pred}} + (1 - w)\, \frac{N_{\text{obs}}}{Y}$$

Term	Meaning
w	Weight given to the model's prediction, between 0 and 1. The rest of the weight goes to the crash record.
0.6	Over-dispersion parameter: how much real crash counts scatter around their average. An assumed typical value for urban roads.
N_{pred}	Crashes per year the model predicts within 250 m of the blackspot (250 m either side gives MoRTH's 500 m blackspot stretch).
Y	Years of crash record for that site, e.g. 3 for 2016–18.
N_{obs}	Crashes recorded at the site over those Y years. Deaths are used if no crash count was published.
$\frac{N_{\text{obs}}}{Y}$	Recorded crashes per year.
N_{EB}	Empirical Bayes estimate of the true crash rate per year: the blend of model and record.
$1 - w$	Weight given to the crash record.

The blackspot multiplier is $N_{\text{EB}} / N_{\text{pred}}$, kept between 1 and 8. It is then adjusted for how deadly the site's crashes are compared with the 0.33 average, and its effect fades with distance from the site. A short record means w is close to 1 (trust the model); a long, heavy record means w is close to 0 (trust the record). Jambuva bridge ends up at ×11.3.

Slide 9: Traffic-flow and scoring formulas

9.1

BPR travel time

Travel time on a road rises as it fills up.

$$t = t_0 \left[1 + 0.15 \left(\frac{V}{C} \right)^4 \right]$$

Term	Meaning
t	Actual travel time on the road at that hour (seconds).
t_0	Free-flow travel time = length ÷ free-flow speed. Free-flow speed is 95% of the speed limit for cars and 92% for two-wheelers.
V	Traffic volume (vehicles per hour).
C	Road capacity (vehicles per hour). Here C means capacity, not the signal cycle used in 9.2.
$\frac{V}{C}$	How full the road is. In the model: the road class's peak V/C (e.g. 1.0 for primary roads, 0.40 for residential) × that hour's share of peak demand (e.g. 1.0 at 9 am, 0.08 at 2 am).
0.15	BPR constant α: at V/C = 1, travel takes 15% longer than free flow.
4	BPR constant β: delay grows with the fourth power of V/C, so it stays small until the road nears capacity, then rises sharply.

9.2 Webster's signal delay

Average waiting time per vehicle at a traffic signal.

$$d = \frac{C\,(1-\lambda)^2}{2\,(1-\lambda x)}$$

Term	Meaning
d	Average uniform delay per vehicle at the signal (seconds).
C	Signal cycle length (one full red-green round), set to 90 s. Here C means the cycle, not the capacity used in 9.1.
λ (lambda)	Green ratio: the fraction of the cycle that is green for your approach, 0.45.
$1-\lambda$	Fraction of the cycle that is red.
x	Degree of saturation: the approach's V/C, capped at 0.95.
λx	How much of the green time is used up by arriving traffic.
2	Part of Webster's derivation, from averaging delay over vehicles arriving uniformly during the cycle.

Example (x = 0.9): $d = \frac{90\times 0.55^2}{2\,(1-0.405)} \approx 23$ s. The model applies this delay in proportion to how likely the junction is to be signalised. Unsignalised junctions instead get a fixed turning delay: 0 s straight, 3 s left, 7 s right, 15 s U-turn.

9.3 Time-of-day risk

Risk per vehicle-km varies with the time of day.

$$\mathrm{TOD}(h) = \frac{\text{share of crashes in } h}{\text{share of traffic in } h}$$

Term	Meaning
$\mathrm{TOD}(h)$	Risk per vehicle-km in window h, compared with the daily average.
h	The 3-hour window containing the departure time (00–03, 03–06, … 21–24).
Share of crashes in h	From MoRTH 2022, Table 7.3 (e.g. 20.4% of crashes between 18:00 and 21:00).
Share of traffic in h	Percentage of the day's driving done in that window. This is an **assumed** typical urban profile, not measured data.

Values are scaled so the traffic-weighted daily average is exactly 1. Result: about 1.3× from midnight to 6 am, about 0.8× from 9 am to noon.

9.4 Pedestrian fatality (Rosén & Sander, 2009)

Chance that a pedestrian struck by a car dies.

$$P(v) = \frac{1}{1 + e^{\,6.9 - 0.09\,v}}$$

Term	Meaning
$P(v)$	Probability that a pedestrian hit at speed v dies.
v	Impact speed in km/h. The model uses the road's traffic speed.
e	Euler's number, ≈ 2.718 (base of natural logarithms).
6.9, 0.09	Constants fitted to real crash data by Rosén & Sander.
$\frac{1}{1+e^{(\cdot)}}$	The logistic S-curve: low at low speeds, rising steeply, levelling off near 1.

Values: 1.5% at 30 km/h, 3.6% at 40, 8.3% at 50, 18% at 60.

9.5 Calibration to MoRTH totals

Turns relative risk into real crashes per year.

$$k = \frac{N_{\text{MoRTH}}}{\sum_e \text{exposure}_e}$$

Term	Meaning
k	One scaling constant per city.
N_{MoRTH}	The city's official yearly crashes from MoRTH 2022 (e.g. Surat 683). The Ahmedabad region uses 1,711 × 1.10 ≈ 1,882 to cover Gandhinagar.
$\sum_e$	Sum over every road link e in the city (junctions are added the same way).
exposure_e	Link e's relative crash potential = road-class crash frequency × road-feature factors × length (km) × daily traffic for that class × 365 ÷ 1,000,000.

Junctions are set to carry 35% of the city's crashes. With k applied, the model predicts exactly as many crashes per year as MoRTH reports.

9.6

Safety score

A 0–100 index shown on every route card.

$$\text{Score} = 100 \, \exp\left(- \ln \frac{100}{70} \cdot \frac{r}{\bar{r}} \right)$$

Term	Meaning
Score	Route safety from 0 to 100; higher is safer.
100	The maximum score, for a route with zero risk.
$\exp(\cdot)$	e raised to the power inside the brackets.
$\ln \frac{100}{70}$	Natural log of 100/70 ≈ 0.357. This constant makes an average road score exactly 70.
r	The route's risk per km = total route risk ÷ route length.
$\bar{r}$ (r-bar)	Average risk per km across the city's roads, for the same mode and hour, weighted by traffic.
$\frac{r}{\bar{r}}$	How risky the route is compared with an average road: 1 = average, 2 = twice as risky.

Check: r/r̄ = 0 gives 100; r/r̄ = 1 gives 70; r/r̄ = 2 gives 49.

Slide 11: Route cost

11

Route cost (what the A* search minimises)

Balances risk against time, set by the user's safety priority.

$$C = \lambda \frac{R}{R_0} + (1 - \lambda) \frac{T}{T_0}$$

Term	Meaning
C	Cost of a candidate route (unit-free). The A* search finds the route with the lowest C. Not the signal cycle from 9.2.
λ (lambda)	The user's safety priority from the slider: 0 = only speed, 1 = only safety, 0.7 by default. Lowered automatically if the safe route would be more than 25% slower (detour cap).
R	Total risk of the candidate route: expected harm from crashes, summed over roads and junction turns, including lighting and blackspot factors and a night-time personal-security term.
R_0	Total risk of the fastest route between the same two points.
$\frac{R}{R_0}$	Risk as a fraction of the normal route's risk: 0.7 means 30% less risk.
T	Total travel time of the candidate route: BPR road times plus junction delays.
T_0	Travel time of the fastest route.
$\frac{T}{T_0}$	Time as a fraction of the normal trip: 1.1 means 10% longer.
$1 - \lambda$	Weight on time.

Dividing by R_0 and T_0 makes risk and time comparable even though they have different units. At $\lambda = 0.5$, one percent extra time is worth exactly one percent less risk.